TransLink Transit Intelligence Pipeline

INFO 4190- PROJECT INTEGRATION 1 | PROJECT PROGRESS REPORT 1

Group 9

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Project Title

TransLink Transit Intelligence Pipeline: A Data Engineering and ML System for Transit Schedule Analysis and Commuter Departure Insights

Team Information

Name	Student ID	Role
Inderpreet Kaur	100383066	Technical Design Lead, system architecture, data pipeline design, database schema, ML model design, API design
Drishti Kapoor	100435109	Design & Documentation Lead, UI/UX wireframes, mockups, requirements specification, literature review, progress reports, final report writing

Current Scope and Objectives

The TransLink Transit Intelligence Pipeline is designed for Metro Vancouver daily commuters, transit researchers, and open data enthusiasts. The core problem it solves is simple: TransLink publishes its full weekly transit schedule as open data, but no public tool currently helps commuters answer the question *"when should I travel?"*, only *"when is the next bus?"*

The system addresses this by processing TransLink's GTFS Static data through a four-layer pipeline: automated data ingestion, ETL processing into PostgreSQL, machine learning analysis using K-Means clustering, and a five-page interactive web dashboard built with Flask and Chart.js.

What the current scope includes:

- GTFS Import Pipeline: automated download and loading of TransLink's weekly data into PostgreSQL
- Route Analyser: shows scheduled service frequency by route, day, time window, and direction
- Stop Explorer: lets users search a stop and view its scheduled service patterns
- Dashboard Filters: filter by route, stop, day of week, time range, transit mode, and direction
- Service Intensity Charts: visualize scheduled frequency patterns
- Plain-Language Summaries: user-friendly explanation of what the chart means, using "scheduled service frequency" and "service intensity", never implying actual crowding or ridership
- Data Refresh Status: shows when the GTFS feed was last processed

K-Means clustering, nearby stop search via PostGIS, route comparison, and export features remain in scope as optional stretch features, implemented only if time permits after core features are complete.

What has changed since the proposal and why:

Three adjustments have been made based on professor feedback received after the first submission:

1. **K-Means repositioned as optional.** The original proposal positioned K-Means clustering as a central component of the system. Based on feedback, K-Means has been moved to an optional stretch feature. The core system will first deliver schedule-frequency aggregation by route, stop, day, time, and direction, which is the genuine commuter value this project provides.
2. **Language standardized around schedule frequency.** The proposal used terms like "busy," "demand," and "congestion" loosely. These have been replaced throughout all design documents with "scheduled service frequency" and "service intensity." The system uses GTFS schedule data as a proxy for service intensity, it cannot confirm actual passenger demand or crowding, and all UI copy and documentation will reflect this clearly.
3. **User testing added to success metrics.** The original proposal lacked user testing criteria. The updated plan includes testing whether users can correctly interpret the plain-language departure recommendation without any instructor explanation, confirming the dashboard communicates effectively to non-technical commuters.

Identified Risks and Challenges

Risk	Description	Impact	Mitigation Strategy
Design scope exceeding timeline	The system includes five dashboard pages. Trip Planner Lite is the most complex and lowest priority page.	If all five pages are pursued equally, the design quality of core pages may suffer.	Trip Planner Lite is formally designated lowest priority. If time is limited, the three core pages, Network Overview, Route Analyser, and Network Patterns, constitute a complete and presentable design on their own.
K-Means producing uninterpretable clusters	K-Means is an unsupervised algorithm, there is no guarantee the clusters it produces will be meaningful or explainable to commuters.	Weak clusters would undermine the Network Patterns dashboard page and reduce the ML component's value.	Design specifies elbow method and silhouette scoring to select optimal k. A rule-based fallback, classifying routes by type and average daily trips, is documented as an

			alternative if clustering results are not meaningful.
GTFS feed format change mid-project	TransLink updates its GTFS Static feed weekly. While the format has been stable, a structural change mid-semester could require minor adjustments to some design documents.	Minor rework on affected design artefacts.	The Week 1 GTFS snapshot is saved and pinned as the reference dataset for all design work this semester. Any feed changes will be assessed at the time and documented as a known variation.
Users misreading schedule frequency as actual crowding	The system uses scheduled service frequency as a proxy for service intensity. Users may incorrectly interpret results as confirming a bus will be crowded.	Could undermine trust in the system if commuters act on misunderstood information.	All UI copy, plain-language summaries, and design documents consistently use "scheduled service frequency" and "service intensity." A visible disclaimer will appear on every dashboard page clarifying that the system reflects schedule data only, not real-time ridership.
Team coordination producing inconsistent design artefacts	Inderpreet and Drishti work in separate technical and documentation roles. Without a clear review process, artefacts could use inconsistent terminology or structure.	Inconsistent documents would reduce the professionalism and coherence of the final design package.	All documents are stored in a shared OneDrive folder using version naming (V1, V2, etc.) rather than overwriting. Drishti holds final ownership of all documentation. Every artefact is reviewed by both team members before submission.

Early Findings

Data Validation – Week 1 Proof of Concept

Before the proposal was submitted, a Python script was written and run against the actual TransLink GTFS dataset to confirm the data was real, accessible, and usable. The script loaded the four most important files, routes, trips, calendar, and stop_times, and returned clean results with no errors. Key confirmed numbers: 242 routes across the full Metro Vancouver network, 63,250 scheduled trips in one week, and 190 R6 Sunday trips alone. The stop_times file came in at 87.78 MB, large but manageable using Pandas column-selective loading. This validated the feasibility of using the TransLink GTFS dataset for the proposed system design.

GTFS File Structure Analysis - Week 4/5 Findings

Following the initial validation, a second and more thorough script was written and run against all 16 files in the TransLink GTFS dataset. This analysis mapped every file's columns, confirmed how files link to each other, and identified what each relationship means for the system design. The key relationships confirmed are:

- routes links to trips via route_id, how the system knows which trips belong to which route
- calendar links to trips via service_id, how the system filters trips by weekday, Saturday, or Sunday
- trips links to stop_times via trip_id, the core chain that powers the Route Analyser feature
- stop_times links to stops via stop_id, connects arrival times to physical stop locations with GPS coordinates
- trips links to shapes via shape_id, provides GPS path data for drawing routes on a map

Two findings from this analysis were not anticipated at the proposal stage:

First, both trips and stops files contain accessibility columns, wheelchair_accessible at the trip level and wheelchair_boarding at the stop level. These were not part of the original design plan but are already in the dataset. This is a meaningful discovery because it means the system can highlight accessibility-friendly trips and stops without needing any additional data source, directly supporting the use case identified through stakeholder feedback below.

Second, the route_color column in routes.txt, which the proposal planned to use for displaying official TransLink brand colours on the dashboard, is empty for regular bus routes. Brand colours appear to be present only for RapidBus and SkyTrain routes. The dashboard design will handle this gracefully by displaying a default colour for routes where no brand colour is available.

Competitive Analysis

The three tools reviewed—TransLink Trip Planner, Google Maps Transit, and the Transit App—were all evaluated based on three years of personal daily use in Metro Vancouver. Each of these tools performs well at answering a simple and important question: *"When is the next bus?"* They provide real-time updates such as delays, vehicle locations, and step-by-step navigation.

However, none of these tools help answer another practical question that many commuters have: *"When is the best time to travel?"* People who are travelling for leisure, running errands, visiting family, or using mobility aids often have flexibility in their departure time and may prefer less crowded or more comfortable travel periods. Despite this, current tools do not help users identify quieter service windows along a route.

This is the gap that the Route Analyser is designed to address by focusing on scheduled service frequency to help users choose better travel times.

Stakeholder Insight

An informal conversation with a regular TransLink commuter highlighted a specific use case that strengthened the project's direction: passengers travelling with strollers, mobility aids, or accompanying elderly family members often prefer to avoid peak service windows simply because quieter trips are more comfortable and less stressful. This confirmed that the plain-language departure suggestion feature has real value beyond general convenience. It also made the accessibility columns discovered in the Week 4/5 file analysis even more relevant, the data to support this user group is already in the dataset.

Preliminary Requirement Refinement

Since the original proposal was submitted, five refinements have been made based on professor feedback. None of these change the core direction of the project, they sharpen and clarify what was already there.

K-Means Clustering- Role Clarified

- **Original:** K-Means was presented as a central component of the system, listed alongside core technologies in the methodology section
- **Updated:** K-Means is now classified as an optional stretch feature
- **Reason:** The core value of this system is schedule-frequency aggregation by route, stop, day, time, and direction, this works entirely without clustering. K-Means adds an extra layer of pattern analysis on top but is not required for the system to function and deliver commuter value. The K-Means model design document remains a full deliverable this semester as part of the complete design blueprint. The optional classification applies only to its implementation priority next semester, core features are built first, and K-Means is implemented if time permits.

User Testing- Added to Success Metrics

- **Original:** No user testing criteria were included in the proposal
- **Updated:** User testing has been added as a formal success metric for the implementation phase
- **Approach:** Test whether a commuter can read a plain-language departure recommendation on the dashboard and correctly understand it, without any explanation from the development team. This confirms the system communicates effectively to non-technical users. Testing will involve 8 participants, a mix of technical users, regular TransLink commuters, and non-technical older adults. Each participant will be shown a screenshot of the plain-language summary cold, with no explanation, and asked: 'What is this telling you and what would you do based on it?' A pass is recorded if the participant correctly identifies both the peak service window and the recommended action. The feature will be considered successful if 6 out of 8 participants pass.

Implementation Roadmap Made Specific

- **Original:** The proposal described a general implementation plan without a clear build sequence
- **Updated:** A specific three-layer roadmap has been defined for next semester

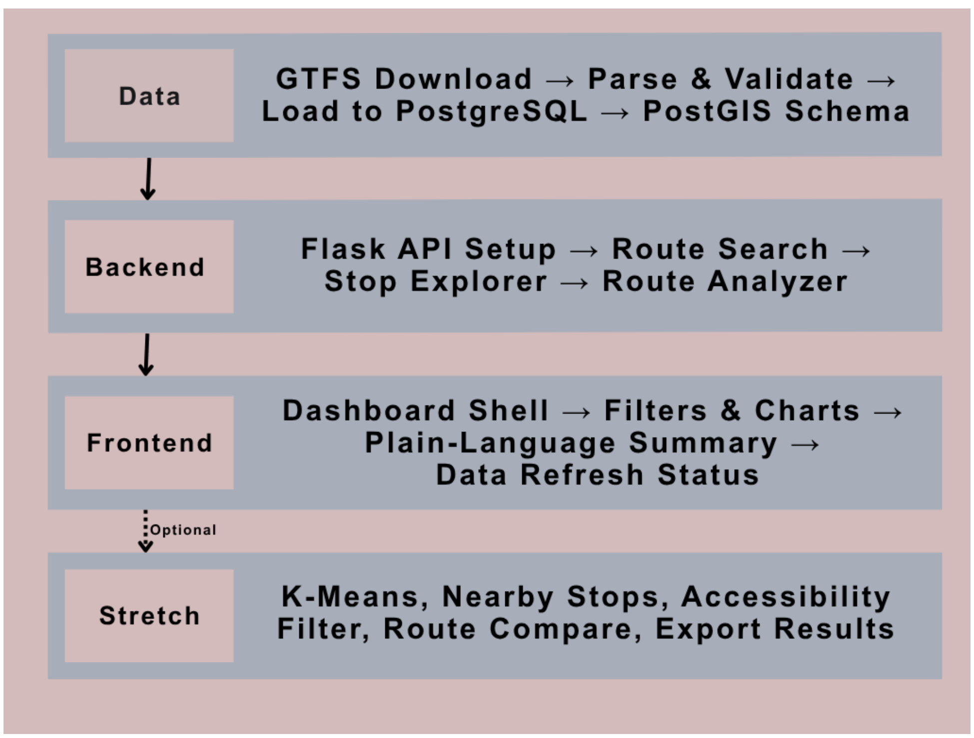

Figure 1: Implementation Roadmap

The roadmap follows this order:

- **Data layer first**: GTFS download, parsing and validation, loading into PostgreSQL, and PostGIS schema setup if stretch features are pursued
- **Backend layer second**: Flask API setup, then Route Search, Stop Explorer, and Route Analyzer endpoints built in sequence
- **Frontend layer third**: Dashboard shell, filters and charts, plain-language summary, and data refresh status display
- **Stretch features last**: K-Means clustering, nearby stop search, accessibility-aware trip filtering, route comparison, and export results, only if time permits after core features are complete

New Stretch Feature: Accessibility-Aware Trip Filtering

- **Original:** Accessibility was not mentioned as a feature
- **Updated:** Added as a new stretch feature following Week 4/5 GTFS file analysis
- **Reason:** The wheelchair_accessible column in trips.txt and wheelchair_boarding column in stops.txt are already present in the TransLink dataset. This feature would allow commuters travelling with strollers, mobility aids, or elderly family members to filter results by accessible trips and stops, directly supporting a user need identified through informal stakeholder feedback

Next Steps

Week 6- Peer Review 1

- Receive another team's Progress Report 1 and complete the peer review form
- Review their scope, risks, research findings, and next steps using the structured template
- Provide constructive, specific, and professional feedback across all five evaluation criteria
- Submit completed peer review form before the Week 6 deadline

Week 6- Early Design Work

- Produce a rough draft of the system architecture diagram showing how the ETL pipeline, PostgreSQL database, Flask API, and Chart.js dashboard connect (Inderpreet leading)
- Produce initial wireframe sketches for Network Overview and Route Analyser pages only, the two highest priority pages (Drishti leading)
- Inderpreet to lead system architecture diagram; Drishti to lead wireframe sketches

Week 7- Midterm Report Preparation

- Begin drafting the Midterm Report structure based on the provided template
- Write the updated Requirements Summary, minimum 10 functional and 5 non-functional requirements in table format
- Expand the Literature Review section with IEEE-style inline citations
- Document the Functional Analysis, compare the planned system against TransLink Trip Planner, Google Maps, and Transit App
- Prepare the Response to Peer Review section based on feedback received in Week 6
- Inderpreet to lead technical sections (requirements, functional analysis); Drishti to lead documentation sections (literature review, formatting, report structure)

Appendix

Appendix A: Week 1 Data Validation Output

A Python script was written and run against the TransLink GTFS dataset before the proposal was submitted to confirm the data was real, accessible, and usable. The validation was performed on the May 12, 2026, snapshot of the TransLink GTFS feed. TransLink publishes an updated feed every Friday, the actual system next semester will automatically download and process the latest weekly feed rather than using this fixed snapshot. The results below confirm the data structure is stable and suitable for the system design.

File	Size	Result
routes.txt	0.05 MB	Found
trips.txt	3.98 MB	Found
calendar.txt	0.03 MB	Found
stop_times.txt	87.78 MB	Found

Key results confirmed:

- Total routes found: **242**
- Total service patterns found: **25**
- Service patterns running on Sundays: **2**
- Total scheduled trips found: **63,250**
- R6 (Scott Road) trips on Sundays: **190**

All four files loaded cleanly with no errors. Column structures were confirmed across all files including route_id, service_id, trip_id, stop_id, arrival_time, and departure_time. The stop_times.txt file at 87.78 MB was the largest file, this confirmed the need for Pandas column-selective loading during implementation to avoid loading the full file into memory unnecessarily.

Appendix B: Full GTFS File Structure Analysis (Week 4/5)

A second script was run against all 16 files in the TransLink GTFS dataset. The table below shows each file, its columns, and its role in the system:

File	Key Columns	Role in System
routes.txt	route_id, route_short_name, route_long_name, route_type	Identifies all 242 routes by name and type

trips.txt	trip_id, route_id, service_id, direction_id, wheelchair_accessible	Links trips to routes and service patterns; includes accessibility data
calendar.txt	service_id, monday–sunday, start_date, end_date	small file (negligible size), contains service calendar definitions
calendar_dates.txt	service_id, date, exception_type	Handles holiday and exception service days
stop_times.txt	trip_id, stop_id, arrival_time, departure_time, stop_sequence	Core file, every scheduled arrival and departure across the network
stops.txt	stop_id, stop_name, stop_lat, stop_lon, wheelchair_boarding	Physical stop locations with GPS coordinates and accessibility data
shapes.txt	shape_id, shape_pt_lat, shape_pt_lon	GPS path data for drawing routes on a map
transfers.txt	from_stop_id, to_stop_id, from_trip_id, to_trip_id	Transfer connections between stops and trips
agency.txt	agency_id, agency_name, agency_timezone	Confirms TransLink as the data publisher
feed_info.txt	feed_start_date, feed_end_date, feed_version	Confirms data validity period: April 20 – June 7, 2026
directions.txt	route_id, direction_id, direction	Compass direction per route (e.g. EAST, NORTH), supports direction filtering
direction_names_exceptions.txt	route_name, direction_id, direction_name	Human-readable direction names per route

route_names_exceptions.txt	route_id, route_name	Override names for specific routes
stop_order_exceptions.txt	route_name, stop_id, stop_name, stop_do	Custom stop ordering for specific routes (e.g. Canada Line)
signup_periods.txt	sign_id, from_date, to_date	TransLink schedule signup periods
translations.txt	table_name, field_name, language, translation	Indigenous language translations of stop names

Three notable findings from this analysis:

- **Accessibility data is already in the dataset** - the wheelchair accessible column in trips.txt and wheelchair boarding column in stops.txt are populated. This directly supports a user need identified through informal stakeholder feedback, a conversation with a family member who regularly commutes on TransLink highlighted that passengers travelling with strollers, wheelchair users, and people accompanying elderly family members often look for quieter, more accessible trips. Since this data already exists in the GTFS feed, an accessibility-aware trip filter has been added as a stretch feature requiring no additional data source.
- **Brand colours are missing for regular bus routes** - the route_color column in routes.txt is empty for standard bus routes. Colours appear only for RapidBus and SkyTrain routes. The dashboard will use a default colour scheme for routes where no brand colour is available.
- **TransLink includes non-standard GTFS files** - directions.txt, direction_names_exceptions.txt, stop_order_exceptions.txt, and translations.txt are TransLink-specific extensions beyond the standard GTFS specification. The directions.txt file provides compass direction per route which directly supports the direction filtering feature in the Route Analyser.

Appendix C: Implementation Roadmap Diagram

The roadmap shows the planned build sequence for next semester across three layers, Data, Backend, and Frontend, followed by optional stretch features. The dashed arrow to the Stretch Features row indicates these are implemented only if time permits after core features are complete.